miRNA Mimic Transfection

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SUSTech-BIO 2026 iGEM Team

Purpose

Introduce synthetic miRNA mimics into adherent cells using Lipofectamine RNAiMAX for miRNA functional studies. The reference setup uses the existing HEK293T / miR-124-3p mimic workflow: 30 nM mimic in 1 mL final volume per 12-well plate well. [1]

The 30 nM working concentration matches GenePharma's general mimic starting guidance. Cell condition, lipid dose and RNA dose still require assessment for each cell line. [4]

Materials & Equipment

Item	Supplier / identifier	Use / specification
miR-124-3p mimic and matched NC mimic	GenePharma; hsa-miR-124-3p and matched mimic NC	Lyophilized mimic / NC, 2 OD per tube in the supplied order; prepare 20 μM stocks [5]
Lipofectamine RNAiMAX	Invitrogen; RNAiMAX product family [2]	Small-RNA transfection reagent; store at 4 °C without freezing
Opti-MEM Reduced Serum Medium	Gibco	Serum-free dilution medium for the two complex-preparation tubes
DEPC-treated water	Sangon Biotech	RNase-free water for oligonucleotide reconstitution and stock dilution
Growth medium for HEK293T	Gibco DMEM + Cellmax FBS	Use the established C01 formulation without added antibiotics during this adapted transfection workflow
12-well culture plate; RNase-free tubes and tips	Axygen tips; laboratory cultureware	Tissue-culture-treated plate; separate tubes for target and control complexes
Pipettes; centrifuge; microscope	Eppendorf centrifuge; laboratory equipment	Liquid handling, brief tube spin and morphology / confluence checks
Cell culture cabinet; CO2 incubator	Laboratory equipment	Aseptic handling; 37 °C and 5% CO2 for routine HEK293T culture

Before You Start

Plan target mimic and a matched non-targeting mimic control at equal RNA concentration and lipid dose. Include untreated cells and a reagent-only mock when assessing baseline health and lipid effects. Identify wells before preparing complexes.

Target healthy, attached HEK293T cells at approximately 60–70% confluence, as in the earlier workflow. Seed in advance using C01/C04 and the observed growth rate. This target falls within both suppliers' confluence guidance. [1, 2, 4]

Check the oligonucleotide identity, amount, diluent and labeled stock concentration. Use the delivered nmol amount or the supplier's conversion for that exact duplex. Do not treat OD as a universal molar unit.

Prepare serum-free Opti-MEM for complex formation. Receiving wells may contain serum. Antibiotic-free growth medium follows the earlier project workflow; RNAiMAX itself does not universally require antibiotics to be excluded. [1, 2]

Procedure

A. Prepare the oligonucleotide stocks

1. Centrifuge the unopened lyophilized mimic tube below 4,000 × g for 4–5 min to collect the powder. Reconstitute target mimic and NC separately with DEPC-treated water. Pipette gently to dissolve, then briefly spin down. [5]

For GenePharma double-stranded mimics, the manual specifies 125 µL DEPC-treated water per OD to make a nominal 20 µM stock: a 2 OD tube therefore receives 250 µL. If the batch certificate specifies the amount in nmol, calculate volume (µL) = nmol × 1000 / 20 and follow the batch-specific value. This OD conversion does not apply to single-stranded products. [3, 5]

2. Label stocks with identity, 20 µM concentration and preparation date. Divide into approximately 10–15 µL working aliquots, seal securely and store at -80°C. Minimize repeated freeze-thaw cycles; the supplied manual recommends using dissolved aliquots within 3 months. [1, 5]

B. Calculate each formulation

3. For each target or control formulation, calculate all ingredients from the same number of well-equivalents. Use 20 µM = 20 pmol/µL; 30 pmol divided by 1 mL final culture volume gives 30 nM. [1]

Component / stage	One 12-well well	Six wells + 10% excess
Tube A: 20 µM mimic or NC	1.5 µL (30 pmol)	9.9 µL (198 pmol)
Tube A: Opti-MEM	48.5 µL	320.1 µL
Tube B: RNAiMAX	3.0 µL	19.8 µL
Tube B: Opti-MEM	47.0 µL	310.2 µL
Combined A + B	100 µL	660 µL; dispense 100 µL/well
Growth medium already in each well	900 µL	900 µL per well

Each A or B tube totals 50 µL per well, or 330 µL for six wells plus excess. Prepare separate mixtures for target and NC. For N wells with 10% excess, multiply every single-well complex ingredient by N × 1.10. [1]

The 3 µL RNAiMAX dose is the earlier 12-well adaptation of the manufacturer's 1.5 µL per 24-well starting amount. Keep lipid dose and RNA dose separately adjustable during optimization. [1, 2]

C. Form and add the complexes

4. Inspect attachment, confluence and morphology. Adjust each receiving well to 900 µL of fresh growth medium, keeping the composition identical across groups. [1]

5. Prepare Tube A by diluting the mimic or NC in Opti-MEM to the calculated total volume. Prepare Tube B separately by diluting RNAiMAX in Opti-MEM. Mix gently. [1]

6. Add Tube A to its corresponding Tube B, mix gently and incubate for 5 min at room temperature. Start timing after the solutions have been combined. [1]

7. Add 100 µL of the appropriate complex mixture to each well. Distribute it across the medium, then gently rock the plate forward-backward and left-right to mix. [1]

For a reagent-only mock, replace the 1.5 µL oligonucleotide stock with the same diluent while keeping 3 µL RNAiMAX and 100 µL total complex-preparation volume. Untreated wells receive 1 mL growth medium without lipid or mimic. These controls answer different questions.

D. Incubate and assess

8. Return HEK293T cultures to 37°C and 5% CO2. Record cell morphology around 24 h and collect the planned functional readout within the selected 24–72 h window. Compare all groups at the same elapsed time. [1]

RNAiMAX does not require immediate complex removal. For assays continuing beyond 24 h, GenePharma suggests replacing the medium at about 24 h to support viability. Apply the same schedule to all groups; a second plasmid transfection follows C04 and the experiment-specific design. [2, 4]

9. Assess cell health and a target-relevant functional readout against the matched NC. For mature-miRNA measurement, use small-RNA-preserving extraction such as MR01 and an appropriate miRNA RT assay; interpret abundance together with function.

p.s.

Count the 100 µL complex mixture within the 1 mL final volume. Adding it to an existing 1 mL of medium produces 1.1 mL and approximately 27.3 nM, not 30 nM.

Opti-MEM in the recipe is added to a defined total: 48.5 µL plus 1.5 µL RNA, and 47 µL plus 3 µL lipid. Adding 50 µL Opti-MEM to each tube would change the dispensing volumes.

Use the matched mimic NC. The delivery also lists a separate 0.50 OD “Negative control”; its function is not established by that label, so it is not substituted for the 2 OD mimic NC. [5]

Use functional evidence to judge miRNA activity. A stronger miRNA qPCR signal or uptake of a fluorescent control alone does not establish the intended target effect.

For experiments involving a second transfection, monitor the combined stress and include controls that receive the same treatment schedule. A broad decrease in cell health or signal cannot be assigned specifically to an MRE.

Expected Outcome

Cells should remain suitable for the planned assay, with morphology and viability documented alongside target and control identities, RNA dose, lipid dose and timing. A mimic effect is assessed relative to its matched NC and an appropriate functional control; no fixed suppression percentage is assumed. [4]

For an MRE reporter experiment, compare the mimic effect on constructs with and without the relevant MRE. This comparison helps distinguish sequence-dependent regulation from general treatment effects.

Troubleshooting

Observed problem	Check first	Action
Rounding, detachment or poor viability	Untreated, mock and NC cultures; input density and medium	If mock is also affected, reassess lipid dose and culture condition. If the effect tracks the target mimic, assess RNA dose and biology with matched controls before attributing mechanism.
Little or no functional effect	Product identity, stock calculation, timing and assay responsiveness	Verify the target and NC labels, then test delivery and a responsive functional control. Optimize RNA and lipid doses with viability readouts.
Large variation among wells	Cell distribution, final volumes, dispensing and timing	Use a shared mixture for each condition, ensure equal complex volumes and distribute complexes consistently. Keep independent culture wells distinct from technical assay repeats.
NC changes the readout	NC chemistry, mock behavior and downstream assay	Confirm a matched NC and compare with untreated/mock wells. Evaluate sequence-independent treatment effects before using the NC as the sole baseline.
miRNA signal increases but target readout is unchanged	Assay specificity, target responsiveness and endpoint timing	Do not equate measured abundance with functional activity. Check a target-responsive assay and the relevant biological context.

Reference

[1] Project protocols: miR-124-3p mimic 12-well transfection and GFAMRE M2 sequential transfection v1.4, sections D2 and F1–F4. GFA+MRE/M2/MRE/. Reopened 2026-10-03.

[2] Thermo Fisher Scientific. Lipofectamine RNAiMAX Reagent Protocol, MAN0007825, Rev. 1.0 (2013), pp. 1–2. Product handling, confluence, complex preparation and per-well lipid dose. Accessed 2026-10-03. Open source

[3] GenePharma. microRNA mimics Product Manual, pp. 2–4; revision/date not printed. Double-stranded product handling, OD-based reconstitution and storage. Accessed 2026-10-03. Open source

[4] GenePharma. miRNA Mimics Usage: transfection starting points and optimization. 12-well dose / final volume, confluence guidance and 24 h medium replacement. Accessed 2026-10-03. Open source

[5] GenePharma. Supplied miRNA manual B072-V001-20240913, pp. 2–4; delivery note dated 17 August 2026. User-supplied photographs reviewed 2026-10-03.